

SMART INDIA HACKATHON 2026

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TITLE PAGE

• Problem Statement ID –	26068
• Problem Statement Title –	WeatherGPT: Conversational AI for Weather Forecasting, Alerts, and Climate Information
• Theme –	Disaster Management
• PS Category –	Software
• Organization –	Ministry of Earth Sciences (MoES) / India Meteorological Department (IMD)
• Team ID –	SIH26068
• Team Name (Registered on portal) –	MEGHDOOT
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MEGHDOOT

IDEA TITLE: WeatherGPT

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'V Proposed Solution (Describe your Idea/Solution/Prototype)

- **Detailed explanation of the proposed solution:**

WeatherGPT is an intelligent conversational meteorological intelligence platform that ingests real-time observation telemetry (AWS), Numerical Weather Prediction models (GFS 0.25° & WRF), and disaster feeds (CAP v1.2). It translates dense atmospheric physics into actionable, contextual natural language decisions with full voice input/output support in 8+ Indian regional languages.

- **How it addresses the problem:**

1. Eliminates Portal Fragmentation: Consolidates separate IMD bulletins, radar layers, and PDF advisories into one conversational interface.
2. Overcomes Jargon: Rather than reporting cryptic millibar pressure or synoptic maps, answers direct citizen questions (e.g. "Can I spray my cotton crop today?").
3. Bridges Rural & Language Divide: Speech-to-text (STT) and text-to-speech (TTS) in native Indian tongues empower non-literate farmers.
4. Accelerates Early Warning: Common Alerting Protocol (CAP) delivers instantaneous Red/Orange/Yellow disaster bulletins to frontline responders.

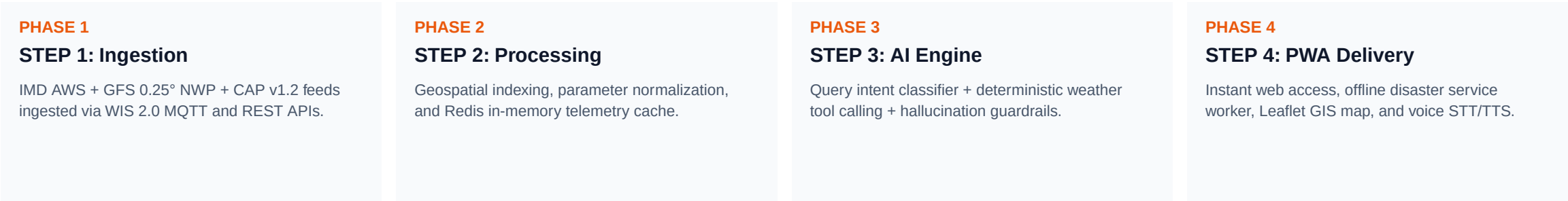
- **Innovation and uniqueness of the solution:**

- Web App with PWA Capability: Chosen over closed native apps to cover a broader space across all devices (Android, iOS, PC, kiosks) with <2MB footprint, zero app-store download hurdle, and offline caching during disaster network blackouts.
- Grounded AI Engine (Zero Hallucinations): Combines conversational LLMs with deterministic meteorological tool calling and atmospheric physics validation.
- Voice-First Rural Access: Built-in regional speech recognition and audio synthesis enabling hands-free use in the field.

• **Technologies to be used (programming languages, frameworks, hardware):**

- Frontend & Client: React 19, TypeScript, Vite, Tailwind CSS, Leaflet GIS Interactive Map, Recharts Climate Visualizer, Web Speech API (Voice STT/TTS).
- Backend & APIs: Node.js, FastAPI, Express, OpenAPI, Zod validation, PostgreSQL with Drizzle ORM, in-memory caching layer.
- AI & Ingestion: LLM Function Calling, Domain Prompt Guardrails, WIS 2.0 / MQTT, GFS 0.25° NWP, WRF models, CAP v1.2 disaster feeds.
- DevOps & Scale: Docker containerization, Kubernetes cluster auto-scaling ready, Edge CDN caching, Progressive Web App (Service Worker).

• **Methodology and process for implementation (Flow Charts / Images / Working Prototype):**



' **Operational Working Prototype Deployed Today:**

Weather Desk (/), Ask WeatherGPT (/ask), Alert Center (/alerts), Signal Map (/map), Climate Lens (/climate), Field Advisory (/advisory).

Repository: <https://github.com/satitushar26-tech/WeatherGPT>

• **Analysis of the feasibility of the idea:**

- Technical Feasibility: Built on production-ready modern stacks (React 19, FastAPI/Node.js, PostgreSQL) with operational prototype.
- Operational Feasibility: Conforms to WMO WIS 2.0 & IMD standard open data formats; integrates seamlessly with government disaster networks.
- Economic Feasibility: Open-source stack eliminates licensing costs; stateless microservices ensure low cloud computing expenditure.
- Adoption Feasibility: PWA architecture eliminates the 50MB app-store download hurdle — accessible immediately via a simple URL link.

• **Potential challenges and risks & Strategies for overcoming them:**

1. AI Hallucination	LLMs hallucinating weather values could endanger public safety.	Strict deterministic tool calling; temperature=0; physical parameter boundary validation against IMD observations.
2. Network Collapse	Cyclones & floods destroy mobile towers, severing connectivity.	PWA Service Worker caches disaster survival guides and last forecast; planned LoRaWAN emergency broadcast mesh.
3. Linguistic Diversity	Complex rural dialects with high voice accent variability.	Web Speech API with regional dialect acoustic models; multi-turn prompt normalization into standardized weather intents.
4. Concurrency Spikes	Millions of concurrent queries hitting servers during cyclones.	Stateless FastAPI/Node.js microservices auto-scaled on Kubernetes; Edge CDN caching on district weather data tiles.

• **Potential impact on the target audience:**

- Ø<ß> Farmers & Rural Communities (700M+ Indians): Crop-weather advisories in regional voice; optimal pesticide spraying, sowing, and irrigation windows reducing crop losses.
- Ø=þ'' Disaster Teams (NDRF / SDMA / DDMA): Early dissemination of CAP disaster bulletins with rapid evacuation checklists and hourly inundation forecasts.
- ' ' þ Aviation & Transport Logistics: Automated METAR/TAF translation into plain language briefings, crosswind components, and highway visibility warnings.
- Ø<ßÜþ Smart Cities & Policy Researchers: Urban heat island mapping, stormwater drainage warnings, and decadal climate anomaly trend analytics.

• **Benefits of the solution (social, economic, environmental, etc.):**

Social Benefits

- Saves lives and prevents injuries through timely early warnings.
- Empowers non-literate and rural citizens through voice accessibility.
- Reduces panic with clear, actionable evacuation guidance.

Economic Benefits

- Protects agricultural yields (mitigates ~150,000+ Cr annual weather crop damage).
- Reduces commercial aviation flight diversions and fuel burn.
- Minimizes municipal property damage from localized urban flash flooding.

Environmental Benefits

- Enables sustainable groundwater and irrigation resource management.
- Tracks decadal temperature & monsoon onset shifts for climate adaptation.
- Facilitates science-driven disaster resilience policies.

- **Details / Links of the reference and research work:**

India Meteorological Department (IMD) - NWP Modeling

Documentation and operational runs of GFS (0.25°) and WRF mesoscale numerical weather forecasting.

URL: <https://mausam.imd.gov.in>

World Meteorological Organization (WMO) - WIS 2.0 Standard

WMO Information System 2.0 architecture and MQTT pub/sub data exchange telemetry standards.

URL: <https://wmo.int>

OASIS Common Alerting Protocol (CAP v1.2)

International public emergency warning format for multi-hazard disaster dissemination.

URL: <https://docs.oasis-open.org/emergency/cap/v1.2>

Ministry of Earth Sciences (MoES) - National Monsoon Mission

High Performance Computing (HPC) Pratyush & Mihir climate modeling frameworks.

URL: <https://moes.gov.in>

National Disaster Management Authority (NDMA) Guidelines

Standard Operating Procedures for Cyclone, Flood, and Disaster Evacuation Planning.

URL: <https://ndma.gov.in>

WeatherGPT Operational Prototype Repository

Source code, PWA service worker, conversational RAG engine, Leaflet GIS map, and voice STT/TTS.

URL: <https://github.com/satitushar26-tech/WeatherGPT>